Question 1

$$\begin{cases} 4x_1 & +bx_2 & +bx_3 & +x_4 & = 4a \\ 3x_1 & +bx_2 & +bx_3 & +x_4 & = 3a \\ -x_1 & +3bx_2 & +3ax_3 & +ax_4 & = -3 \\ -2x_1 & +3bx_2 & +3ax_3 & +(2a-3)x_4 & = -6 \\ 7x_1 & +6bx_2 & +(3a+3b)x_3 & +2ax_4 & = 9a-6 \end{cases}$$

- $R_1 \leftarrow R_1 - R_2$:

$$(4x_1 - 3x_1) + 0x_2 + 0x_3 + 0x_4 = 4a - 3a \implies x_1 = a$$

- Eliminate x_1:

$$\begin{aligned} R_2 \leftarrow R_2 - 3R_1 : &\quad 0x_1 + bx_2 + bx_3 + x_4 = 0 \\ R_3 \leftarrow R_3 - 3R_1 : &\quad 0x_1 + 3bx_2 + 3ax_3 + ax_4 = a - 3 \\ R_4 \leftarrow R_4 - 3R_1 : &\quad 0x_1 + 3bx_2 + 3ax_3 + (2a-3)x_4 = 2a - 6 \\ R_5 \leftarrow R_5 - 6R_1 : &\quad 0x_1 + 6bx_2 + (3a+3b)x_3 + 2ax_4 = 2a - 6 \end{aligned}$$

- $R_4 \leftarrow R_4 - R_3$:

$$0x_2 + 0x_3 + (a-3)x_4 = a - 3 \implies (a-3)x_4 = a - 3$$

  - If $a \neq 3$: $x_4 = 1$.
  - If $a = 3$: The equation becomes $0 = 0$ (redundant).

(a) No Solution

Condition: $a \neq 3$ and $b = 0$.
Substituting $b = 0$ into Equation 2 gives $0 = -1$, which is impossible.

$$\boxed{a \neq 3 \text{ and } b = 0}$$

(b) Unique Solution

Condition: $a \neq 3$, $b \neq 0$, and $b \neq a$.
Unique Solution:

$$x_1 = a, \quad x_2 = -\frac{1}{b}, \quad x_3 = 0, \quad x_4 = 1$$

$$\boxed{a \neq 3,\ b \neq 0,\ \text{and } b \neq a}$$

(c) Infinitely Many Solutions (1 Parameter)

Conditions:

1. $a \neq 3$ and $b = a$, or
2. $a = 3$ and $b \neq 3$.

General Solutions:

- **Case 1** ($a \neq 3, b = a$)**:**

$$x_1 = a, \quad x_2 = t, \quad x_3 = -\frac{1}{a} - t, \quad x_4 = 1 \quad (t \in \mathbb{R})$$

- **Case 2** ($a = 3, b \neq 3$)**:**

$$x_1 = 3, \quad x_2 = t, \quad x_3 = 0, \quad x_4 = -bt \quad (t \in \mathbb{R})$$

$$\boxed{(a \neq 3 \text{ and } b = a) \text{ or } (a = 3 \text{ and } b \neq 3)}$$

(d) Infinitely Many Solutions (2 Parameters)

Condition: $a = 3$ and $b = 3$.
General Solution:

$$x_1 = 3, \quad x_2 = s, \quad x_3 = t, \quad x_4 = -3s - 3t \quad (s, t \in \mathbb{R})$$

$$\boxed{a = 3 \text{ and } b = 3}$$

Question 2

(a) Linear System

The pumping rates equations:

$$\begin{cases} (2x + 3y + 4z) \times 7{,}200\,\mathrm{s} = 2{,}880{,}000 \\ (x + 2y + z) \times 14{,}400\,\mathrm{s} = 2{,}880{,}000 \\ (x + y + 3z) \times 14{,}400\,\mathrm{s} = 2{,}880{,}000 \end{cases}$$

simplified:

$$\begin{cases} 2x + 3y + 4z = 400 \\ x + 2y + z = 200 \\ x + y + 3z = 200 \end{cases}$$

(b) Existence of Unique Solution

The coefficient matrix is:

$$\begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

Calculating its determinant:

$$\begin{aligned} \det &= 2 \cdot \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} - 3 \cdot \begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix} + 4 \cdot \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} \\ &= 2 \cdot (2 \cdot 3 - 1 \cdot 1) - 3 \cdot (1 \cdot 3 - 1 \cdot 1) + 4 \cdot (1 \cdot 1 - 2 \cdot 1) \\ &= 2 \cdot 5 - 3 \cdot 2 + 4 \cdot (-1) \\ &= 10 - 6 - 4 = 0. \end{aligned}$$

Since the determinant is zero, the system is linearly dependent. To solve:

1. Subtract Equation 3 from Equation 2:

$$(x + 2y + z) - (x + y + 3z) = 200 - 200 \implies y - 2z = 0 \implies \boxed{y = 2z}.$$

2. Substitute $y = 2z$ into Equation 2:

$$x + 2(2z) + z = 200 \implies x + 5z = 200 \implies \boxed{x = 200 - 5z}.$$

3. Check:

   - Substitute $x = 200 - 5z$ and $y = 2z$ into Equation 1:

$$\begin{aligned} 2(200 - 5z) + 3(2z) + 4z = 400 &\implies 400 - 10z + 6z + 4z = 400 \\ &\implies 400 = 400 \quad \text{(True)}. \end{aligned}$$

- Substitute into Equation 3:

$$(200 - 5z) + 2z + 3z = 200 \implies 200 = 200 \quad \text{(True)}.$$

The system has infinitely many solutions parameterized by z. Thus, there is insufficient information for unique determination:

$$\boxed{\text{No, the system has infinitely many solutions.}}$$

(c) Number of Pumps of Type X

Given $z = 10\, l/s$, substitute into the reduced equations:

$$\begin{cases} x = 200 - 5z \\ y = 2z \end{cases}$$

$$y = 2(10) = 20\, l/s, \quad x = 200 - 5(10) = 150\, l/s.$$

The required total flow rate to fill the pool in 1 hour (3600 seconds) is:

$$\frac{2{,}880{,}000\,\mathrm{L}}{3600\,\mathrm{s}} = 800\, l/s.$$

Let k be the number of type X pumps. The equation becomes:

$$150k + 2(20) + 10 = 800 \implies 150k = 750 \implies k = 5.$$

$$\boxed{5}$$

Question 3

$$\begin{cases} x^2 + 4xy + 4xz + 5y^2 + 4yz + 5z^2 = 22 \\ x^2 - xy - xz + 5y^2 + 2yz + 4z^2 = 29 \\ 5x^2 + 0xy + 3xz + y^2 + 2yz + 3z^2 = 16 \\ -x^2 + xy + 2xz + 3y^2 + 5yz + 0z^2 = 7 \\ 5x^2 + 3xy + 4xz + 0y^2 + 3yz + 0z^2 = 0 \\ 5x^2 - xy + 4xz - y^2 + 3yz + 5z^2 = 23 \end{cases}$$

Solution

Let:

$$\begin{cases} a = x^2, \\ b = xy, \\ c = xz, \\ d = y^2, \\ e = yz, \\ f = z^2. \end{cases}$$

The system becomes:

$$\begin{cases} a + 4b + 4c + 5d + 4e + 5f = 22 \quad (1) \\ a - b - c + 5d + 2e + 4f = 29 \quad (2) \\ 5a + 3c + d + 2e + 3f = 16 \quad (3) \\ -a + b + 2c + 3d + 5e = 7 \quad (4) \\ 5a + 3b + 4c + 3e = 0 \quad (5) \\ 5a - b + 4c - d + 3e + 5f = 23 \quad (6) \end{cases}$$

Gaussian Elimination

- Eliminate a:
  - $R_2 \leftarrow R_2 - R_1$:
  $$0a - 5b - 5c + 0d - 2e - f = 7$$
  - $R_3 \leftarrow R_3 - 5R_1$:
  $$0a - 20b - 17c - 24d - 18e - 22f = -94$$
  - $R_4 \leftarrow R_4 + R_1$:
  $$0a + 5b + 6c + 8d + 9e + 5f = 29$$
  - $R_5 \leftarrow R_5 - 5R_1$:
  $$0a - 17b - 16c - 25d - 17e - 25f = -110$$

- $R_6 \leftarrow R_6 - 5R_1$:

$$0a - 21b - 16c - 26d - 17e - 20f = -87$$

Becomes:

$$\left[\begin{array}{cccccc|c} 1 & 4 & 4 & 5 & 4 & 5 & 22 \\ 0 & -5 & -5 & 0 & -2 & -1 & 7 \\ 0 & -20 & -17 & -24 & -18 & -22 & -94 \\ 0 & 5 & 6 & 8 & 9 & 5 & 29 \\ 0 & -17 & -16 & -25 & -17 & -25 & -110 \\ 0 & -21 & -16 & -26 & -17 & -20 & -87 \end{array}\right]$$

- Pivot on b in R_2:
  1. Multiply R_2 by -1: $5b + 5c + 2e + f = -7$.
  2. Eliminate b from $R_3 - R_6$ using R_2:
     - $R_3 \leftarrow R_3 + 4R_2$:
     $$0b + 3c - 24d - 10e - 18f = -122$$
     - $R_4 \leftarrow R_4 - R_2$:
     $$0b + c + 8d + 7e + 4f = 36$$
     - $R_5 \leftarrow R_5 + \frac{17}{5}R_2$:
     $$0b + c - 25d - \frac{51}{5}e - \frac{108}{5}f = -\frac{849}{5}$$
     - $R_6 \leftarrow R_6 + \frac{21}{5}R_2$:
     $$0b + 5c - 26d - \frac{43}{5}e - \frac{79}{5}f = -\frac{582}{5}$$

  Becomes:

$$\left[\begin{array}{cccccc|c} 1 & 4 & 4 & 5 & 4 & 5 & 22 \\ 0 & 5 & 5 & 0 & 2 & 1 & -7 \\ 0 & 0 & 3 & -24 & -10 & -18 & -122 \\ 0 & 0 & 1 & 8 & 7 & 4 & 36 \\ 0 & 0 & 1 & -25 & -\frac{51}{5} & -\frac{108}{5} & -\frac{849}{5} \\ 0 & 0 & 5 & -26 & -\frac{43}{5} & -\frac{79}{5} & -\frac{582}{5} \end{array}\right]$$

- Pivot on c in R_4, swap R_3 and R_4. Eliminate c from R_3, R_5, R_6:
  - $R_3 \leftarrow R_3 - 3R_4$:
  $$0c - 48d - 31e - 30f = -230$$
  - $R_5 \leftarrow R_5 - R_4$:
  $$0c - 33d - \frac{86}{5}e - \frac{128}{5}f = -\frac{849}{5}$$

- $R_6 \leftarrow R_6 - 5R_4$:

$$0c - 66d - \frac{218}{5}e - \frac{179}{5}f = -\frac{1482}{5}$$

Becomes:

$$\begin{bmatrix} 1 & 4 & 4 & 5 & 4 & 5 & | & 22 \\ 0 & 5 & 5 & 0 & 2 & 1 & | & -7 \\ 0 & 0 & 1 & 8 & 7 & 4 & | & 36 \\ 0 & 0 & 0 & -48 & -31 & -30 & | & -230 \\ 0 & 0 & 0 & -33 & -\frac{86}{5} & -\frac{128}{5} & | & -\frac{849}{5} \\ 0 & 0 & 0 & -66 & -\frac{218}{5} & -\frac{179}{5} & | & -\frac{1482}{5} \end{bmatrix}$$

- Pivot on d in R_3:

1. Multiply R_3 by -1: $48d + 31e + 30f = 230$.
2. Eliminate d from R_5, R_6 using R_3:
   - $R_5 \leftarrow R_5 + \frac{33}{48}R_3$:

$$0d + \frac{329}{80}e - \frac{199}{40}f = -\frac{467}{40}$$

   - $R_6 \leftarrow R_6 + \frac{66}{48}R_3$:

$$0d - \frac{39}{40}e + \frac{109}{20}f = \frac{397}{20}$$

3. Multiply R_5 and R_6 by 80:
   - R_5:

$$329e - 398f = -934$$

   - R_6:

$$-39e + 218f = 794$$

Becomes:

$$\begin{bmatrix} 1 & 4 & 4 & 5 & 4 & 5 & | & 22 \\ 0 & 5 & 5 & 0 & 2 & 1 & | & -7 \\ 0 & 0 & 1 & 8 & 7 & 4 & | & 36 \\ 0 & 0 & 0 & 48 & 31 & 30 & | & 230 \\ 0 & 0 & 0 & 0 & 329 & -398 & | & -934 \\ 0 & 0 & 0 & 0 & -39 & 218 & | & 794 \end{bmatrix}$$

- Find f, e, d, c, b, a
  - Use R_6:

$$-39e = 794 - 218f \implies e = \frac{218f - 794}{39}$$

  - Use R_5:

$$329\left(\frac{218f - 794}{39}\right) - 398f = -934$$

$$329(218f - 794) - 398f \cdot 39 = -934 \cdot 39$$

$$71{,}722f - 261{,}326 - 15{,}522f = -36{,}426$$

$$56{,}200f = 224{,}900 \implies f = 4$$

- Use R_6:

$$e = \frac{218(4) - 794}{39} = \frac{78}{39} = 2$$

- Use R_3:

$$48d + 31(2) + 30(4) = 230 \implies 48d = 48 \implies d = 1$$

- Use R_4:

$$c + 8(1) + 7(2) + 4(4) = 36 \implies c = -2$$

- Use R_2:

$$5b + 5(-2) + 2(2) + 4 = -7 \implies 5b = -5 \implies b = -1$$

- Use R_1:

$$a + 4(-1) + 4(-2) + 5(1) + 4(2) + 5(4) = 22 \implies a = 1$$

- Becomes:

$$a = 1, \quad b = -1, \quad c = -2, \quad d = 1, \quad e = 2, \quad f = 4$$

Final Solution

From the values:

$$\begin{cases} a = x^2 = 1 \implies x = \pm 1, \\ b = xy = -1, \\ c = xz = -2, \\ d = y^2 = 1 \implies y = \pm 1, \\ e = yz = 2, \\ f = z^2 = 4 \implies z = \pm 2. \end{cases}$$

- combinations of x, y, z:

$$x \in \{1, -1\}, \\ y \in \{1, -1\}, \\ z \in \{2, -2\}.$$

- Apply constraints $xy = -1$, $xz = -2$, and $yz = 2$:

1. $x = 1$:
   - $xy = -1 \implies y = -1$.
   - $xz = -2 \implies z = -2$.
   - Check $yz = (-1)(-2) = 2$ ✓.

   Solution: $(x, y, z) = (1, -1, -2)$.
2. $x = -1$:
   - $xy = -1 \implies y = 1$.
   - $xz = -2 \implies z = 2$.

  - Check $yz = (1)(2) = 2$ ✓.

  Solution: $(x, y, z) = (-1, 1, 2)$.

- Reject invalid combinations:
  - $x = 1,\ y = 1$: Fails $xy = -1$.
  - $x = -1,\ y = -1$: Fails $xy = -1$.
  - $z = 2$ with $x = 1$: Fails $xz = -2$.
  - $z = -2$ with $x = -1$: Fails $xz = -2$.

$$\boxed{(x, y, z) = (-1, 1, 2) \quad \text{and} \quad (1, -1, -2)}$$

Question 4

It is given that

$$A \xrightarrow{R_3+R_1} \xrightarrow{R_4-R_2} \xrightarrow{R_2+5R_1} \xrightarrow{R_2\leftrightarrow R_3} \xrightarrow{2R_3} \begin{pmatrix} 1 & 2 & a & 3 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & a-1 & 1 \\ 0 & 0 & 0 & a+2 \end{pmatrix}$$

for some constant $a \in \mathbb{R}$.

Solution (a)

Find E_1, E_2, E_3, E_4, E_5, such that

$$A = E_1E_2E_3E_4E_5 \begin{pmatrix} 1 & 2 & a & 3 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & a-1 & 1 \\ 0 & 0 & 0 & a+2 \end{pmatrix},$$

reverse the given row operations in inverse order.

1. - Original: $R_3 \leftarrow R_3 + R_1$
   - Inverse: $R_3 \leftarrow R_3 - R_1$
   - Elementary Matrix E_5:

$$E_5 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

2. - Original: $R_4 \leftarrow R_4 - R_2$
   - Inverse: $R_4 \leftarrow R_4 + R_2$
   - Elementary Matrix E_4:

$$E_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

3. - Original: $R_2 \leftarrow R_2 + 5R_1$
   - Inverse: $R_2 \leftarrow R_2 - 5R_1$

- Elementary Matrix E_3:

$$E_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -5 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

4.
- Original: $R_2 \leftrightarrow R_3$
- Inverse: $R_2 \leftrightarrow R_3$ (swap is self-inverse)
- Elementary Matrix E_2:

$$E_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

5.
- Original: $R_3 \leftarrow 2R_3$
- Inverse: $R_3 \leftarrow \frac{1}{2}R_3$
- Elementary Matrix E_1:

$$E_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Solution (b)

The homogeneous system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions if $\det(A) = 0$.

$$\begin{pmatrix} \boxed{1} & 2 & a & 3 \\ 0 & \boxed{2} & 1 & 3 \\ 0 & 0 & a-1 & 1 \\ 0 & 0 & 0 & a+2 \end{pmatrix}.$$

$$\det(A) = 1 \cdot 2 \cdot (a-1) \cdot (a+2) = 2(a-1)(a+2).$$

Set $\det(A) = 0$:

$$2(a-1)(a+2) = 0.$$

Since $2 \neq 0$, the solutions are:

$$a - 1 = 0 \implies a = 1, \quad \text{or} \quad a + 2 = 0 \implies a = -2.$$

Values of a: $\boxed{a = 1 \text{ or } a = -2}$.